

This document outlines the scope, timeline, and architectural impact of migrating the **attechglobal.com** backend from a hybrid SQLite/JSON file system to a unified **Supabase** infrastructure.

Project Inventory

LAYER	CURRENT TECHNOLOGY	MIGRATION REQUIRED
SQLite Database	sqlite3 (images.db)	Yes (SQL Tables)
JSON Metadata	25+ JSON files in /data/metadata	Yes (Supabase DB)
Authentication	Hardcoded creds + Custom JWT	Yes (Supabase Auth)
Image Storage	Local /public/uploads	Yes (Supabase Storage)
API Layer	Next.js API Routes	Yes (Supabase SDK)

Phased Implementation Plan

PHASE	TASKS & DELIVERABLES	EST. TIME
1. Setup	Project initialization, Schema definition, RLS policies	2 – 3 hrs
2. DB Logic	Replace SQLite helpers in src/lib/database.ts	3 – 4 hrs
3. Metadata	Migrate JSON files to page_metadata table	3 – 4 hrs
4. Auth	Supabase Auth integration & middleware update	2 – 3 hrs

PHASE	TASKS & DELIVERABLES	EST. TIME
5. Storage	Public uploads migration to Supabase Buckets	3 – 4 hrs
6. API Rewrite	Updating all 5 endpoint groups to use SDK	4 – 5 hrs
7. Data Migration	Scripted export/import of current live data	1 – 2 hrs
8. QA	Full system testing & dashboard verification	3 – 4 hrs

Total Estimated Development Time

21 – 29 Hours

Approximately 3–4 days (Full-time) or 7–8 days (Part-time)

 **Risk: Image Migration**

All existing assets in `public/uploads/` must be programmatically re-uploaded to Supabase Storage. Database references must be updated to the new CDN URLs.

 **Structural Change: SEO Management**

The current `fileSeoUtils.ts` writes directly to `.tsx` files. Post-migration, SEO will be served dynamically from the database, improving build times and reliability.

 **Server Compatibility**

Next.js must be configured in SSR mode for API routes to function with Supabase. Verify if the current cPanel hosting supports Node.js runtime environments.

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